Distributed Databases

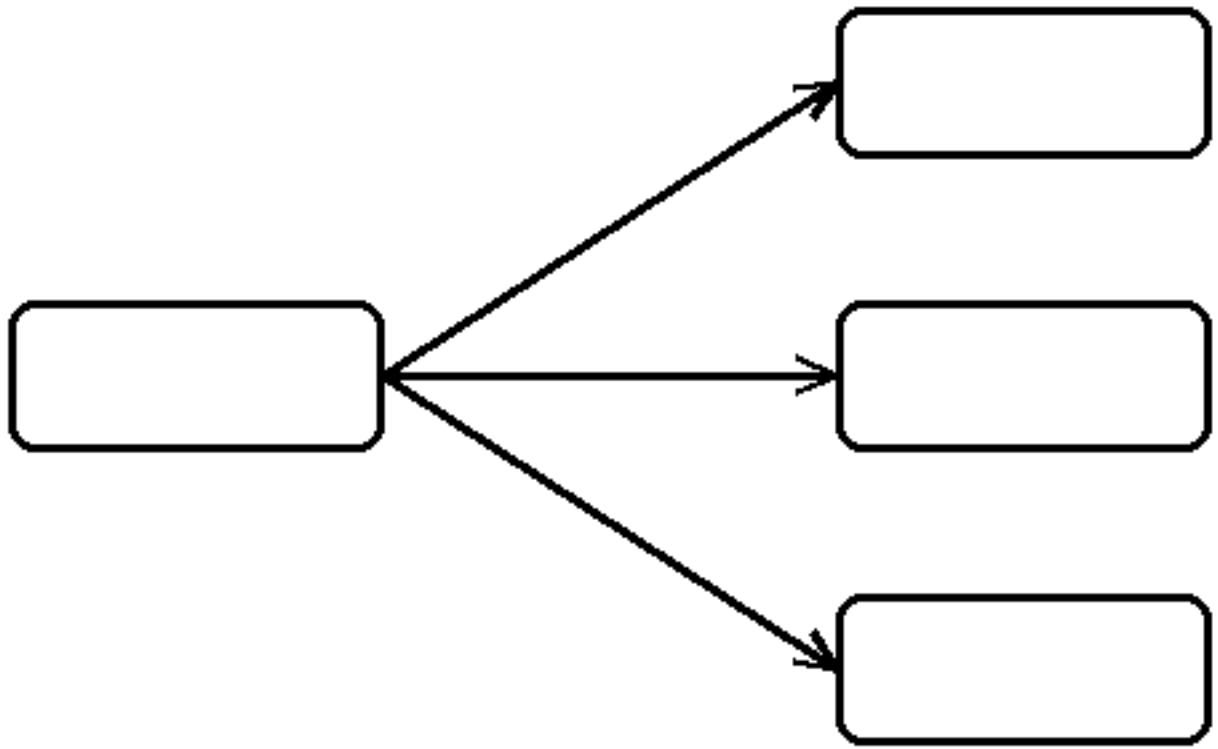
Week 7

Why replicate?

Availability

Latency

Durability



$$R + W > N$$

Quorums

See lecture

Sloppy quorum / hinted handoff

Leader-based

Sync vs async

Replication lag!!

Read-your-writes

Monotonic reads

Multi-leader ? conflicts

LWW

CAP: a system can be consistent, available AND partition-tolerant at the same time

Vector clocks

Version vectors

Anti-entropy

Merkle trees

Consensus

Raft: leader, term, log

Paxos (skip)

2PC

Exam: know these

Summary

Questions?